



Metal-induced structural variability of mononuclear metal-binding sites from a database perspective

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Definitions:

apo-protein: the polypeptide part of a holo-protein (i.e. the holo-protein without its prosthetic group).

donor atom: an atom that forms a direct interaction (coordination bond) with the metal ion.

holo-protein: a protein bound to its prosthetic group.

MBS: metal-binding site.

ABSTRACT

Metalloproteins are ubiquitous in all kingdoms of life. Their role and function are tightly related to the local structure of the metal-binding site. In this regard, the MetalPDB database is an invaluable tool since it stores the 3D structure of metal-binding sites and of their corresponding apo forms. In this work, we exploited MetalPDB to compute extensive statistics over >3000 clusters of mononuclear sites about the rearrangements occurring upon change in metalation state. For each cluster, we matched the holo and apo sites so that it was possible to average the distances between all possible pairs of Ca and donor atoms and thus quantitatively assess structural variations by computing the Δ values (mean apo distance – mean holo distance). **For most of the structures the backbone is rigid with little to no rearrangement, while donor atoms experience significant changes of their relative position when the metal is removed.** Sodium and potassium sites are an exception to this general observation. This is most likely caused by their preference for coordination by the main-chain oxygen atoms, making the rearrangement of donor atoms superimposable to that of the backbone. Magnesium and calcium show a different behavior, despite their chemical similarity: calcium sites undergo a larger reorganization upon metalation although both metals have similar percentage of backbone oxygen as donor atoms. We ascribe this observation to the structural and energetic factors regulating the selectivity for calcium over magnesium.

1. Introduction

38% of the entries in the Protein Data Bank (PDB [1]) harbor one or more metal ions [2,3]. Furthermore, it has been shown that at least 40% of enzymes need metal ions for their biological function [4,5]. Thus, it is not surprising that a variety of metals are essential for organisms in all kingdoms of life [6,7]. However, many essential metals, particularly from the d-block, are poorly available in the environment and can become cytotoxic in high amounts. Consequently, the intracellular concentration of metal ions is strictly controlled [8,9], thanks to the joint action of transport, delivery, storage, detoxification, and efflux machineries. Bacterial pathogens must acquire their required metal ions from the host organism. To counter this, the host can deploy a protective mechanism, called nutritional immunity, aimed at inhibiting the growth of pathogens by restricting the availability of crucial metal ions [10]. A similar strategy can be pursued also through pharmacological treatment [11].

The reactivity and physiological role of metal ions in metalloproteins is largely defined by the local structure environment, which determines how the metal is placed in the active site, how it interacts with the substrate and, for redox-active metals, its reduction potential [12,13]. The metal-binding site (MBS) can be thought as a substructure around the metal ion(s) that represents the macromolecular environment sensed by the metal. This substructure should correspond to the minimal environment determining the function of the metal, i.e., the “minimal functional site” [14]. The MBS can be defined in a way such that it is possible to automatically extract these sites from the 3D structures of metalloproteins available from the PDB. Atomic-level knowledge of the MBSs is crucial to understand the function of these systems. However, sometimes it is not straightforward to obtain a metalloprotein sample of sufficient quality for structural studies. This can be caused by various factors; for example, the metal ion can be lost during purification or never be incorporated into the target protein under the experimental conditions used to produce the protein. In such cases, it may still be

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possible to determine the 3D structure of the protein without its metal cofactor(s), namely of the apo-protein.

The availability of the 3D structure of the apo-protein enables structure-based prediction of the occurrence of metal-sites, by leveraging the information on the relative position in space of the potential donor atoms that coordinate the metal. In addition, the success of AlphaFold and AlphaFold2 in the CASP initiatives [15–18] has made it possible to obtain high-quality computer-generated 3D structural models for proteins not yet characterized experimentally [19–21]. In turn, this may allow novel MBSs to be discovered [22]. Initial attempts in this direction assumed that the side chains of the protein residues were already correctly placed in space. This is very likely when the cofactor is essential for folding, i.e. when in vivo the protein exists only in the holo-form. However, there are numerous cases where both the apo- and holo-proteins may exist in vivo, e.g. in the context of signaling processes or of metal storage. This can be relevant in particular for systems binding individual metal ions or small metal-containing cofactors, such as dimetallic clusters, where the hydrophobic core of the protein is sufficiently large for a compact 3D structure to form [23]. The occurrence of physiologically relevant apo- and holo-protein pairs is important for two distinct reasons. From the structural bioinformatics perspective, it poses the issue of systems where the predicted conformation may not be the correct one to dock the cofactor(s). Secondly, and perhaps more fundamentally, it corresponds to two distinct structural states that interconvert in the cell upon metal binding, with presumably different properties with respect to e.g. the recognition of molecular partners or their thermodynamic and kinetic stability (and ultimately lifetime) in the cell [24,25].

Here we took advantage of the MetalPDB database of MBSs [2,3] to create an extensive dataset of apo- and holo-pairs of mononuclear sites. We used the dataset to compute statistics of MBS rearrangements. We observed that for the majority of physiologically relevant metal ions, there are statistically significant rearrangements of the position of the donor atoms upon change of metallation state. Sodium, potassium and magnesium sites were noticeable exceptions to this general trend. Our results have implications for metalloprotein structure-based prediction, metalloprotein design and the understanding of the fundamental structural properties of metalloproteins.

2. Methods

We used a SQL query to retrieve all data from the MetalPDB database [2,3]. The latter stores a collection of 3D templates automatically extracted from the PDB describing the local environment around metal (s). In the automated procedure, for each metal atom in the structure, the non-hydrogen atoms at a distance smaller than 3.0 Å are identified as its donor atoms (DAs), i.e. the atoms that bind directly to the metal. The protein residues or small molecules that contain at least one donor atom are the metal ligands (called endogenous or exogenous, respectively). The full metal-binding site (MBS hereafter) is obtained by including any other residue or chemical species having at least one atom within 5.0 Å from a metal ligand. MetalPDB groups all MBSs in clusters containing both “equistructural” and “equivalent” sites [2,3]. For the construction of our dataset, we retrieved the clusters containing only “equivalent” sites and their corresponding apo structures. Two MBSs are defined as equivalent in MetalPDB if they satisfy all the following conditions: (i) they are found in PDB chains with the same structure (based on Pfam domain composition or on the sequence identity between the two chains being $\geq 50\%$); (ii) after structural superposition of the PDB chains, they are superimposed with the metal atoms in the same position; (iii) they contain the same metal(s). For each cluster we acquired only mononuclear sites sharing the same metal ligands.

In MetalPDB all atoms of a site are already labeled with their structural role (i.e., ligand; ligand neighbor; other). This information was propagated to the apo sites in each cluster through sequence alignment. This allowed us also to properly associate residues across all

the sites in the cluster regardless of possible differences in the numbering schemes of the original structures. After assembling the dataset, we used in-house-written Python scripts to parse the coordinates of the sites. Firstly, we separated the residues based on their protein chain and removed all exogenous ligands (i.e., non-protein, non-nucleic acid). Only the conformation with the highest occupancy was retained for residues having multiple conformations in the PDB file. Subsequently, the program extracts the coordinates and B-factors of the C α and the donor atoms (DAs) for each ligand. These coordinates are used to compute the distances between all pairs of C α atoms and all pairs of donor atoms. The list of all the MetalPDB clusters, labeled by the corresponding database identifier, together with the PDB identifiers of the holo and apo-sites in each cluster is given in the Supplementary material (Table S1).

Since we ensured that all sites had the same ligands and consequently the same atom pairs, we could average the measured distances over all holo or apo sites for each cluster. By subtracting the mean holo distance from the mean apo distance, we computed the Δ value. Negative Δ values correspond to sites that contract upon metal removal, whereas positive Δ values correspond to sites that expand upon metal removal. The absolute value of Δ quantifies the change occurring upon metalation.

Pymol was exploited for structure visualization, as well as for the alignment of the sites to their corresponding structures [26]. The Metals² tool was used for pairwise structural alignment of MBSs [27].

Finally, to assess if the changes in the distributions of the distances between the C α and DA pairs upon metalation, and between their corresponding Δ values were statistically significant, we computed the Wilcoxon signed-rank test [28] using the Python library SciPy [29].

3. Results

Our pairing was based on the clusters of equivalent mononuclear MBSs in MetalPDB. These were further filtered in order to retain sub-clusters that had the same metal-binding pattern. MetalPDB associates to each cluster of MBSs the corresponding apo-structures contained in the PDB on the basis of sequence similarity (with a 50% threshold). For the present analysis we kept only apo sites where all the ligands were conserved and had the same spacing as in the corresponding cluster of holo-MBSs. In practice, we constructed groups of holo-MBSs and of the corresponding apo-sites, all sharing the same metal-binding pattern. We assembled a dataset of 3074 groups, for 30 different metals (Table 1). The number of apo-sites in Table 1 is larger than the holo-sites because MetalPDB automatically applies a redundancy filter to the latter. However, this imbalance did not affect the subsequent analysis, which were all based exclusively on the group averages. Indeed, to evaluate the structural deviations between the holo/apo pairs, we measured all intra-site distances between C α atoms as well as between donor atoms (DAs) and averaged them for holo/apo sites within each group. Consequently, we compared 7042 pairs of distances for the C α and 8607 for the DAs in the 3074 clusters. We did not attempt to separate MBSs on the basis of the oxidation state of the bound metal. This was motivated by the fact that the X-ray radiation applied to protein crystals may cause undetected reduction of the metal sites [30]. This makes it difficult to ascertain the metal oxidation state at the database scale.

Fig. 1 shows the distribution of distances for zinc sites, separated by the number of protein ligands in each site. The corresponding plots for all other metals are reported in the Supplementary Material.

The data of Fig. 1 readily show that the distributions of C α -C α distances are relatively similar for holo- and apo-sites, whereas there are strong differences in the distributions of distances between DAs. This suggests that the structural rearrangements of the protein backbone occurring upon change of the metalation state of a MBS are typically small. On the other hand, significant changes in the position of the side chains appear to be more common in our dataset.

The Wilcoxon signed-rank test was used to reject the null hypothesis

Table 1

Assembled dataset. Each row reports the total number of inspected clusters for the specified metal, with the corresponding number of holo- and apo- sites.

Metal	Holo sites	Apo sites	N° of clusters
As	2	15	1
Ba	17	181	11
Ca	2418	12,829	486
Cd	387	5739	136
Co	67	513	28
Cs	9	978	7
Cu	341	2203	50
Eu	2	7	2
Fe	1270	577	39
Gd	8	39	6
Hg	247	1986	67
K	610	6668	178
Li	11	83	6
Mg	1616	26,277	538
Mn	286	1931	74
Na	2019	28,524	779
Ni	158	1147	71
Pb	8	344	6
Pd	4	356	4
Pr	10	44	7
Pt	28	128	10
Rb	7	896	4
Sm	25	187	14
Sr	21	53	3
Tl	6	16	2
U	52	102	6
Y	10	205	8
Yb	2	54	2
Zn	5440	14,196	529

that the distributions of the C α -C α and DA-DA distances are the same for the apo- and holo-sites [31]. Table 2 includes all the metals for which we detected a significant difference between the apo- and holo-sites at the $p < 10^{-3}$ level. In all cases, the deviation was more significant for the distances between the DAs than between the corresponding C α atoms. Indeed, we observed significant differences for the C α -C α distance distributions only for calcium, magnesium, manganese, sodium and zinc (Table 2). Thus, these five metals experience statistically meaningful rearrangements in the conformation of the backbone of the MBSs upon change of metalation state.

We selected X-ray structures with a resolution in the range 1.5–2.0 Å to investigate whether apo-structures were appreciably “floppier” than the corresponding holo sites. We focused on the 1.5–2.0 resolution range in order to avoid possible biases due to different mathematical treatment of the anisotropy of B-factors. In the selected range all B-factors are considered isotropic, which, also thanks to the relatively small range breadth, should make it possible to compare them in a meaningful way (Fig. 2). Because we requested that both the apo- and holo-structure fell in the desired resolution range, our sample contains significantly less data than the whole dataset. Eventually, we analyzed only the data for four different metals, all having >100 clusters, namely sodium, magnesium, calcium and zinc. Whereas the B-factors of C α and DA atoms are quite similar for the former two metals in both forms, there are noticeable differences for calcium and zinc sites. For the latter, B-factors (and hence flexibility) are higher in the apo- than in the holo-sites. The effect is more pronounced for the DAs, in line with the qualitative notion that metal binding “fixes” the position of residue side chains.

We then decided to evaluate what is the relationship between the rearrangements of the relative position of the C α atoms, which can be used as a proxy for the protein backbone rearrangements, and the corresponding donor atoms. A straightforward way to visualize the different behavior of these two groups is through the frequency count of the deviations measured for all site pairs. The example of zinc sites is displayed in Fig. 3, whereas the corresponding graphs for all other metals are reported in the Supplementary material.

The distributions of changes in the C α -C α distances have sharp peaks

at around 0 Å and are modestly skewed towards positive values (indicating that the site is more open in the apo-structure) with a few sites experiencing rearrangements up to 10 Å. Interestingly, there are some instances of negative values, indicating compaction of the backbone upon metal removal. For the donor atom distances, the peak of the distributions is also at about 0 Å. However, these distributions are broader and highly skewed towards positive values, with changes up to 4 Å being common. Several values larger than 10 Å are observed. Furthermore, we aimed to assess the average effect of metalation over the entire site, rather than on each distance individually, to understand whether the structural rearrangements are spread over all ligands. We did this by analyzing the per-site average Δ values. The Wilcoxon signed-rank test indicated significant differences between the backbone and donor atom rearrangements for 17 metals (Table 3, at $p < 10^{-3}$ level).

Fig. 4 shows the correlation between the distance changes observed for the C α -C α pairs and for the donor atom pairs. For clarity, here we averaged the changes over each entire site, so that any dot in the plot corresponds to a single MBS. Qualitatively, we can identify two different behaviors. When the rearrangement of the protein backbone is very large (of the order of 10 Å or more), there is an essentially linear correlation. This is presumably due to the fact that the overall protein structure changes greatly upon metalation of the site and both the backbone and the side chain are greatly displaced. On the other hand, smaller backbone rearrangements, which are most cases in our dataset, can correspond to a very different extent of side chain rearrangements, going from negligible up to 8–10 Å (Fig. 4, inset). Sites with small rearrangements for both the C α and the donor atoms are pre-organized MBSs, where the coordination environment for the metal ion is structurally defined ahead of metal binding [32]. However, in the majority of cases, the structural rearrangements of the donor atoms were somewhat larger than the corresponding backbone rearrangements. A peculiar case is that of MBSs containing ligands that are consecutive in sequence. In these cases, (red dots in Fig. 4), the C α -C α distance is fixed, and structural rearrangements can occur only for the ligand side chains.

4. Discussion

We examined 3074 clusters of MBSs by evaluating the C α -C α and DA-DA (donor atom) distances for all metal ligands in both holo- and apo-protein structures. Pairings between these structures were taken from the clusters stored in the MetalPDB database. For the present analysis, we took into account only mononuclear sites (i.e. MBSs harboring a single metal ion) where all the ligands as well as the metal identity were conserved in all the members of that cluster.

For most metals there is a statistically significant difference between the distributions of the DA-DA distances in the holo- vs apo-sites, based on a Wilcoxon signed-rank test at $p < 10^{-5}$. These included all biologically relevant metal ions. For some metals (Table 2), also the C α -C α distances were significantly different upon change of metalation state. Instead, there were no metals for which a significant variation was observed only for the C α -C α distances but not for the donor atoms. This is reasonable because a backbone rearrangement almost inevitably causes the side chains to reposition as well. In a previous analysis based on 210 sites, it was observed, in line with our findings here, that rearrangements of the MBSs were relatively common (occurring in about 40% of the dataset) and that such rearrangements typically occurred at the level of the side chains of the metal ligands, even when the backbone conformation was preserved upon metalation [33].

We then looked at the average reorganization of sites, which is possibly more relevant in the perspective of prediction/identification of MBSs from apo-protein structures. This is because sites where the majority of the metal ligands, even if not all, are already structurally pre-disposed for metal binding should be identifiable with relative ease. Instead, sites where all the ligands rearrange upon metalation, and hence are associated with a larger mean variation of distances, are

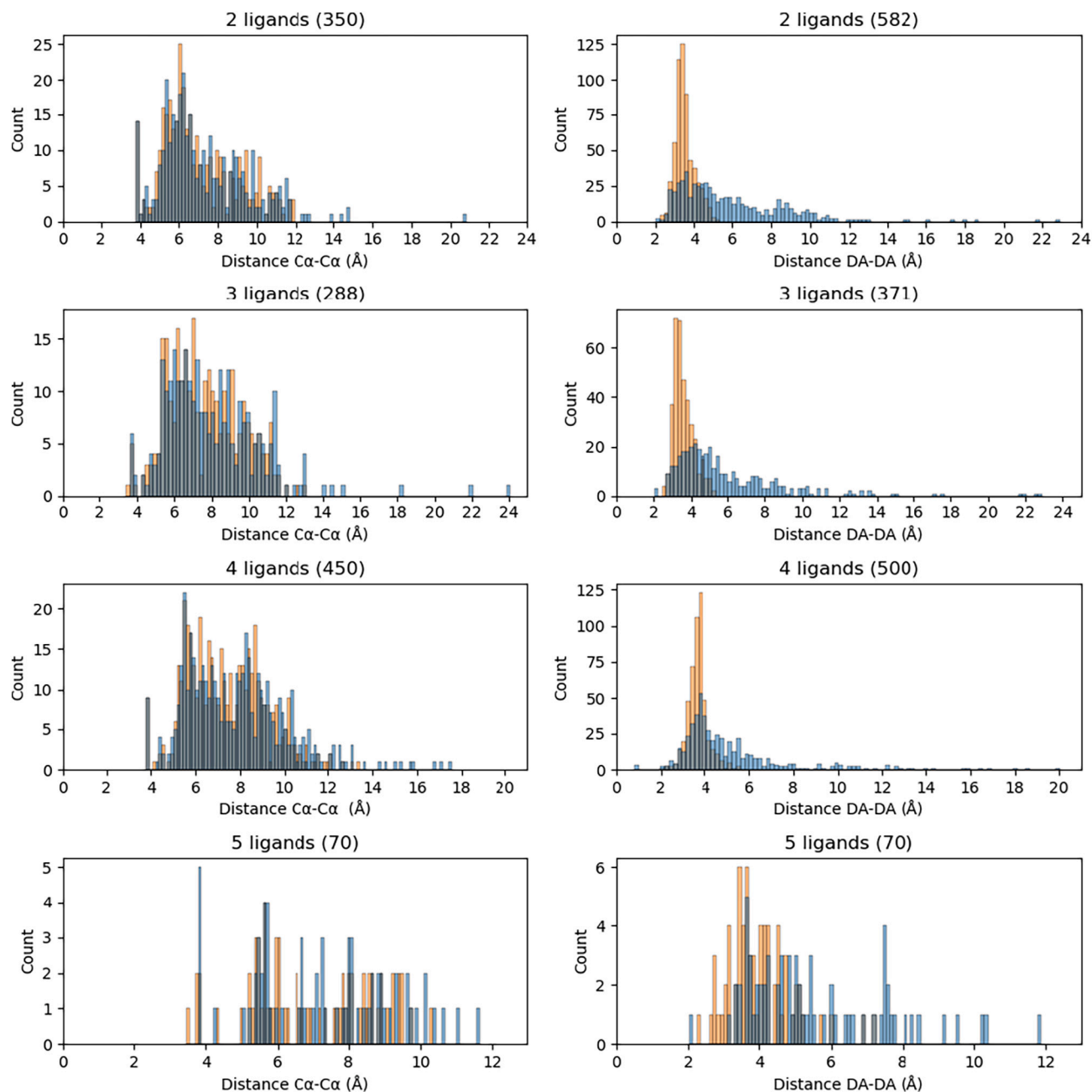


Fig. 1. Frequency counts of Cα-Cα (left) and donor atom (right) distances in zinc sites (orange: holo; blue: apo), separated by the number of endogenous metal ligands. The number of distances included in each panel is also shown. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

presumably harder to identify with e.g. template-based prediction methods. Our results indicate that the average displacement of donor atoms is larger than the average displacement of the backbone. In fact, because our statistics on donor atoms sometimes include backbone oxygen atoms, the conformational reorganization of the side chains of the metal ligands is presumably even larger than suggested by the data in Tables 2 and 3. For all metals the median value of the variations of absolute distance values among Cα atoms are below 0.2 Å, being practically zero for physiological metal ions. For the latter ions, three quarters of the sites have a backbone rearrangement smaller than 0.4 Å, except Fe. Note that we consider absolute distances in order not to worry whether the site contracts or expands upon metalation. Both behaviors should be similarly problematic from the site prediction perspective.

When looking at the corresponding data for donor atoms, we

observed two different behaviors for physiological metal ions. The median ΔDA of transition metals goes from 1.2 to almost 2 Å (Table 3). These metals show preference for side chain coordination, thus metalation results in an appreciable conformational rearrangement of the donor atoms. Instead, the median of the distance variation for potassium and sodium is only about 0.2 Å. To explain the difference from the other metals, we investigated the nature of the donor atoms involved in the coordination of sodium and potassium. They are mainly coordinated by the main-chain oxygen atoms. 82% of the DA-DA distances for sodium sites involve at least one backbone oxygen, whereas the corresponding value for potassium is 84%. The donor atoms thus experience structural rearrangements whose extent is closely comparable to the backbone [34]. This can also justify the very small effect of the change of metalation state on the B-factors of sodium sites (Fig. 2).

Table 2

P-values for the comparison between the distance distributions in apo- vs holo-sites. The *p*-value obtained from the Wilcoxon signed-rank test indicates the likelihood that the compared distributions are the same. The table lists all metals for which at least one of the Cα-Cα or DA-DA distance distributions differed in the comparison at a *p*-value threshold of 10^{-3} . The values in bold highlight the comparisons with a *p*-value $<10^{-5}$.

Metal	<i>p</i> -value for the Cα-Cα distance distribution	<i>p</i> -value for the DA-DA distance distribution	N° of Cα-Cα distances	N° of DA-DA distances
Ca	2.10E-19	3.25E-230	1516	2021
Cd	1.48E-04	8.48E-49	182	344
Co	5.70E-02	2.31E-13	63	89
Cu	1.09E-01	3.68E-21	113	141
Fe	5.93E-04	2.70E-14	81	96
Hg	5.58E-01	8.95E-05	103	121
K	3.03E-02	1.47E-43	553	603
Mg	5.22E-11	1.91E-98	1096	1216
Mn	4.46E-08	9.96E-27	167	202
Na	1.27E-16	4.55E-120	1713	1828
Ni	2.53E-04	3.33E-25	133	164
Pb	1.20E-01	2.94E-04	17	28
Pr	1.29E-01	1.49E-05	9	26
Pt	1.37E-02	9.77E-04	10	11
Sm	2.11E-01	9.09E-08	16	38
Y	5.19E-01	2.38E-07	12	23
Zn	3.30E-13	5.57E-173	1173	1543

The median Δ DA of magnesium (0.38 Å) is like that of sodium and potassium and is quite different from that of calcium (1.03 Å), despite their chemical similarity. Both metal ions have a similar percentage of

backbone oxygens acting as donor atoms (about 60%), and in general share a preference for having oxygen atoms in their first coordination spheres, also from Glx/Asx side chains. A major difference, as already noted in the literature, is that calcium typically involves a larger number of protein residues in its coordination sphere, whereas magnesium involves more water molecules or other exogenous ligands (such as ATP) [35]. However, these considerations do not provide a rationale for the larger extent of DA reorganization in calcium vs magnesium sites. It is possible that this is linked to structural factors underlying the selectivity for calcium over magnesium ions in calcium-binding proteins. Energetic considerations indeed suggest that flexible MBSs should favor calcium binding [36]. In line with this, we indeed observed that the B-factors of calcium sites featured somewhat higher values in the apo-form, indicating that metal binding reduces the flexibility of the site, at variance with what happens for magnesium where the B-factors of apo- and holo-sites are similar (and close to the value for calcium holo-sites). We can speculate that the structural behavior of the calcium sites is instrumental for their functional role in signaling processes [37].

We inspected a selection of structures representing typical patterns of structural changes in our dataset (Figs. 5–7). Cases associated to negative displacements of the Cα atoms upon metal removal (i.e. negative values along the x-axis of Fig. 4) feature either rearrangements of at least one of the fragments harboring a metal ligand, leading the backbone in the apo-structure to occupy the position of the metal, or the rigid shift of one structural element closer to the others. In the latter case, it is likely that the distance among the side chains is also shorter in the apo- vs the holo-structure. An example is the formation of disulfide bridges in the absence of the zinc ion seen in PDB entry 5ax2 (Fig. 5B)

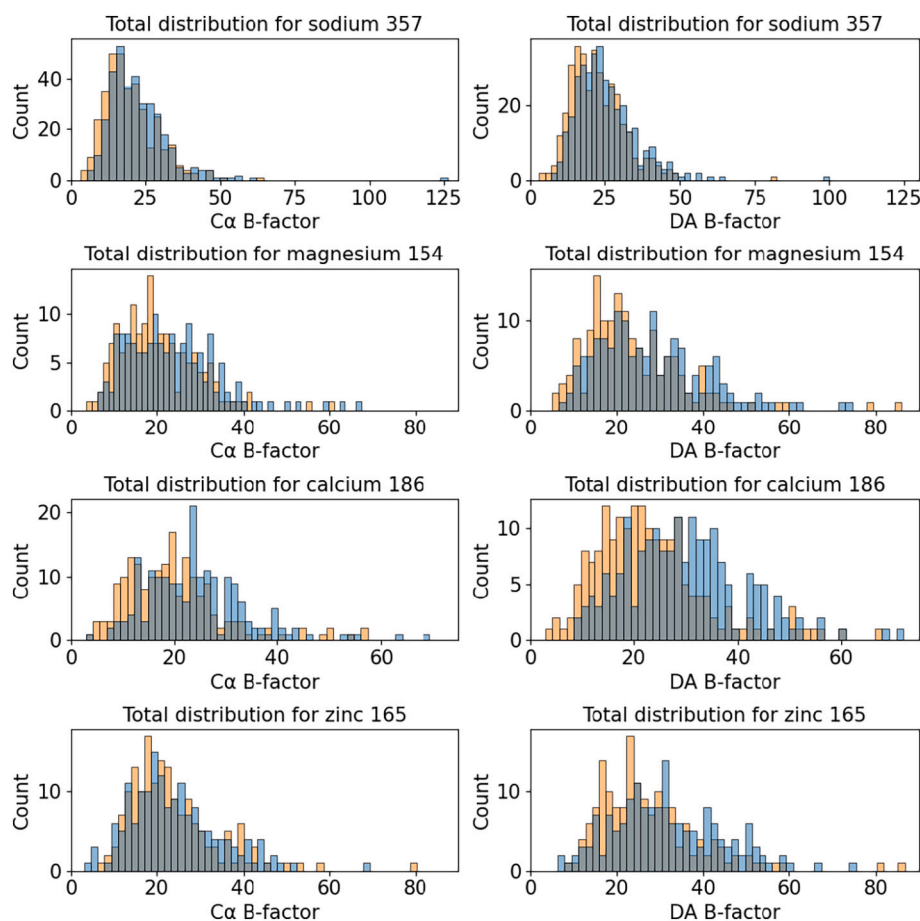


Fig. 2. Frequency count of B-factors for Cα (left) and donor atoms (right). The values were extracted for each atom and averaged over all sites of a cluster. Each panel shows the holo (orange) and apo (blue) distribution for the four metals having >100 clusters with both apo- and holo-sites in the 1.5–2.0 resolution range. In the panel of each metal is reported the count. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

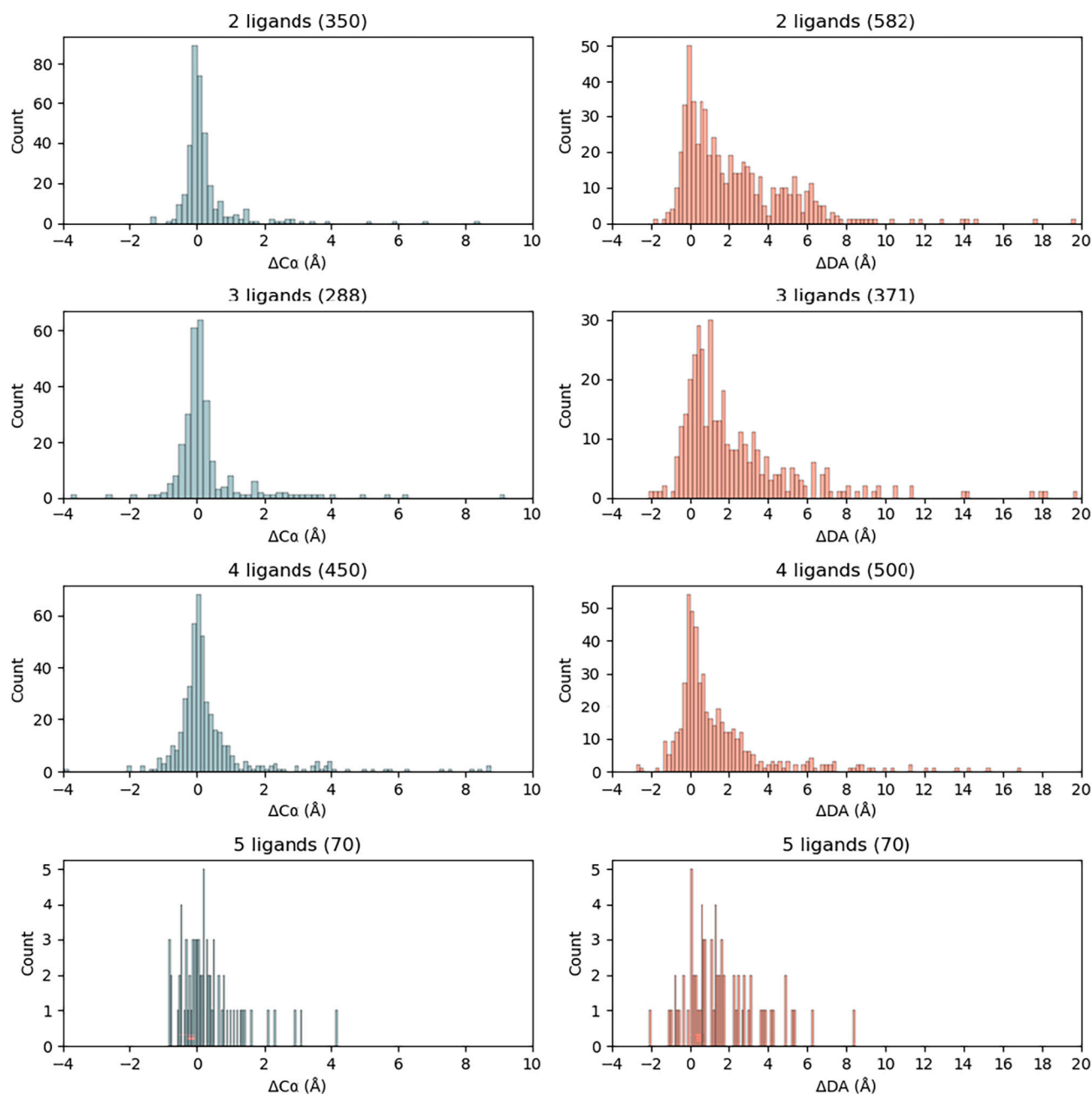


Fig. 3. Frequency counts of apo vs holo structural changes measured for C α -C α (left) and donor atom (right) distances in zinc sites, separated by number of endogenous metal ligands. All changes are measured as apo- minus the corresponding holo- distances.

[38]. In a few cases the displacement of the DAs is larger than the C α atoms. For example, in the HypBA1 hydrolase two of the four ligands of the MBS are displaced in the apo-structure (Fig. 5A), causing a partial unwinding of a helix [39].

Instead, when the $\Delta C\alpha$ values are close to zero and the ΔDA values are of the order of a few Å, it is likely that the local backbone structure, especially when having a high secondary structure content, is maintained and only the rotameric state of the ligand side chains is perturbed (Fig. 6A). Alternatively, sites containing ligands that are consecutive along the sequence or separated by just one residue, tend to have small or null $\Delta C\alpha$ values because of the constraints imposed by the covalent chemical structure of the polypeptide chain, even when the local 3D structure undergoes relevant changes upon change of metallation state (Fig. 6B).

Only in a handful of cases we observed large-scale rearrangements of

the whole protein backbone (Fig. 7). These correspond to cases where the MBS is located at a domain-domain interface, so that metal removal causes the two structural domains to reorient with respect to one another. This is not a general property of interfacial sites, as there are known cases where the interface is stable even in the absence of metal [40].

5. Concluding remarks

This work provides an extensive overview of the structural rearrangements taking place in clusters of mononuclear metal-binding sites. To assemble our dataset we took all metalloprotein clusters already stored in MetalPDB and filtered them to ensure that the apo-/holo-protein pairs analyzed here maintained all the residues in the metal-binding pattern. In this way, we could obtain information also from

Table 3

P-values computed for averaged sites with the Wilcoxon signed-rank test. The median data are computed over the absolute Δ values to reflect the overall behavior of the site upon metalation. The table lists all metals for which at least one of the C α -C α or DA-DA distance distributions differed in the comparison at a *p*-value threshold of 10^{-3} . The most significant values ($p < 10^{-5}$) are highlighted in bold.

Metal	<i>p</i> -value	Median Δ C α (Å)	Median Δ DA (Å)
Ca	4.50E-66	0.02	1.03
Cd	7.38E-19	0.04	1.62
Co	1.08E-04	0.02	1.68
Cu	1.42E-08	0.04	1.54
Fe	2.81E-07	0.09	1.26
Hg	1.87E-03	0.00	0.19
K	5.87E-15	0.01	0.22
Mg	4.92E-51	0.02	0.38
Mn	2.33E-10	0.07	1.22
Na	2.05E-52	0.02	0.18
Ni	1.67E-11	0.04	2.54
Pb	6.25E-02	0.15	3.01
Pr	1.56E-02	0.05	1.91
Pt	3.91E-03	0.16	2.05
Sm	3.66E-04	0.03	2.97
Y	7.81E-03	0.12	3.10
Zn	1.35E-71	0.04	1.30

apo-/holo-protein pairs of homologous proteins from different organisms.

The distributions of C α and DA distances for the metal ligands were statistically different for most metals, and in particular for all the physiological metals. Nevertheless, 75% of the MBSs harboring physiological metals had C α distance variations within 0.2 Å, which means that the backbone conformation was typically maintained upon (de) metalation. On the other hand, there were significant changes in the relative position of DA for all metals except sodium, potassium,

magnesium. For the two alkaline metals, the reason can be attributed to their first coordination sphere involving mainly backbone oxygen atoms (in addition to exogenous ligands). For magnesium, this is not true and we hypothesize that the small DA rearrangements observed here are due to the surface location of magnesium-binding sites and/or to the comparatively small binding energy of the metal. In terms of template-based prediction of MBSs, our findings suggest that backbone-driven methods might perform best.

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CRediT authorship contribution statement

Milana Bazayeva: Data curation, Formal analysis, Writing – original draft. **Vincenzo Laveglia:** Data curation, Methodology. **Claudia Andreini:** Conceptualization, Writing – review & editing. **Antonio Rosato:** Conceptualization, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data are summarized in the Supplementary Table S1 and can be downloaded from the MetalPDB database (<https://metaldpdb.cerm.unifi.it/>)

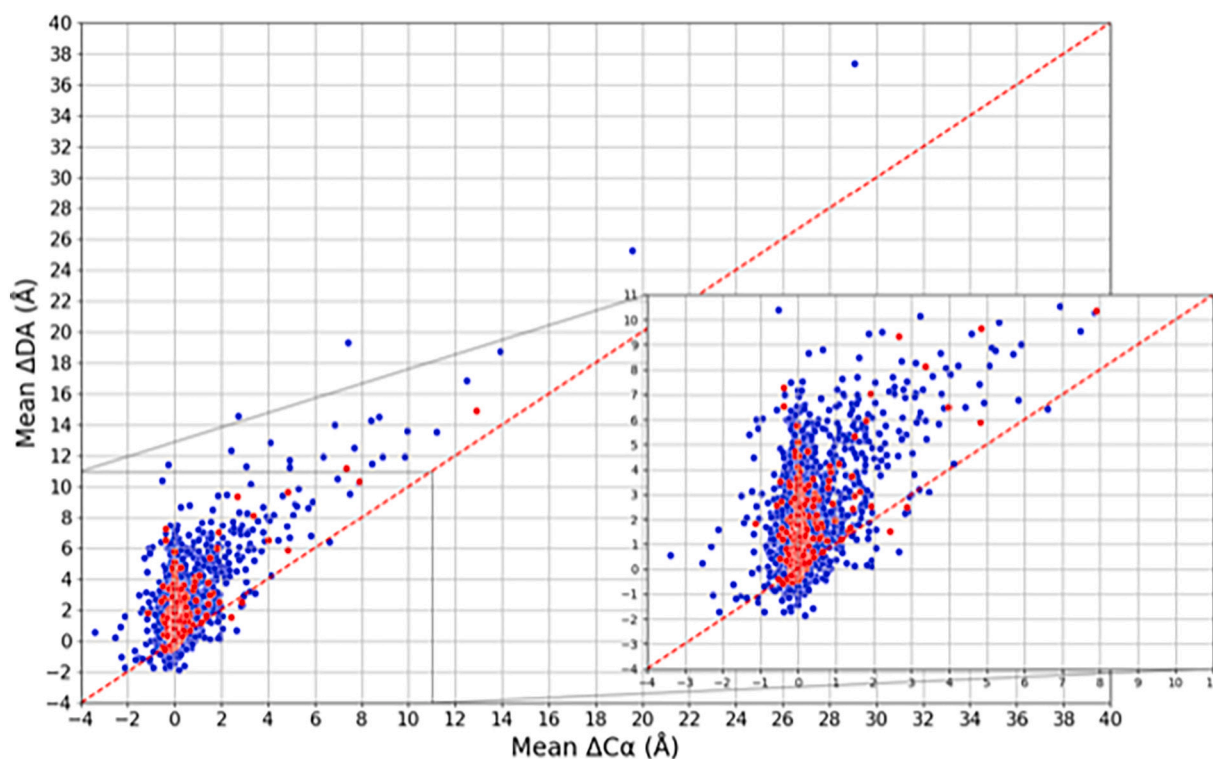


Fig. 4. Mean Δ values over entire MBSs. The red line is $y = x$ and serves only to guide the eye. Red dots correspond to sites having at least two metal ligands that are consecutive in sequence, hence featuring at least one fixed C α -C α distance. All changes are measured as apo- minus the corresponding holo- distances. All the metals of Table 3 are included in the plot. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

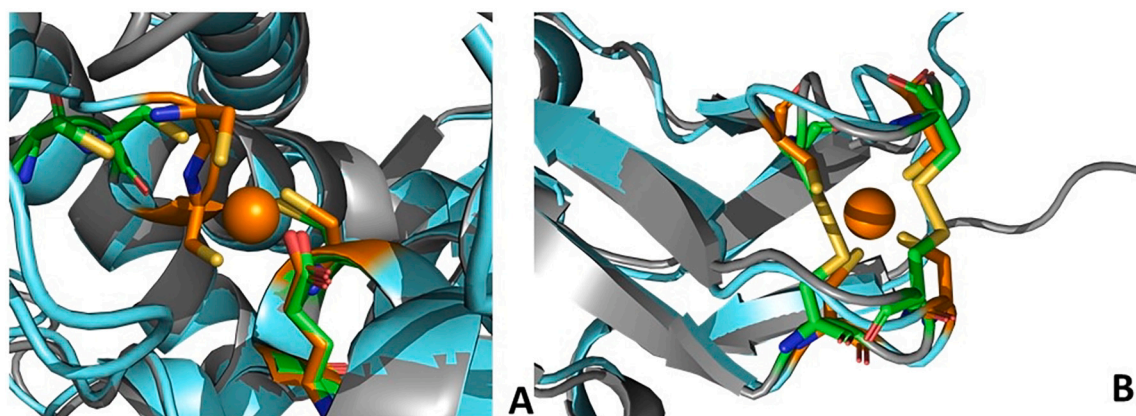


Fig. 5. Representation of selected patterns of structural rearrangements. A. Apo: 3wrf and Holo: 3wks (2.42, 1.52). B. Apo: 5ax2 and Holo: 1wge (-0.24, -0.32). The holo-structure (cyan, metal ligands in orange) is superimposed to the corresponding apo (gray, metal ligands in green). In parenthesis we reported for each panel the coordinates of the corresponding point in Fig. 4. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

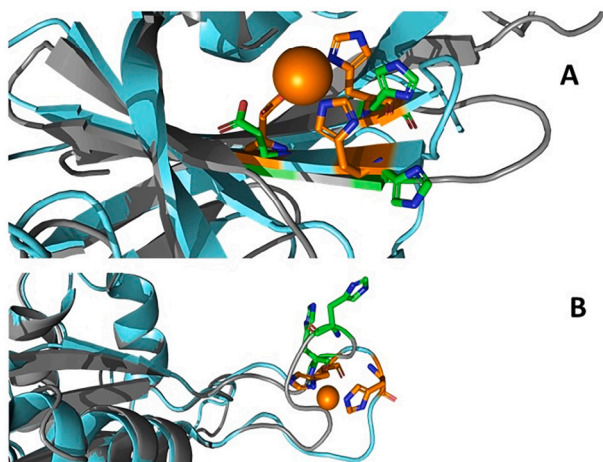


Fig. 6. Examples of large-scale structural rearrangements. A. Apo: 1lj6 and Holo: 3n9b (0.292, 7.07). B. Apo: 3e1u and Holo: 3ir2 (0.40, 5.69). The colour code is as in Fig. 4.

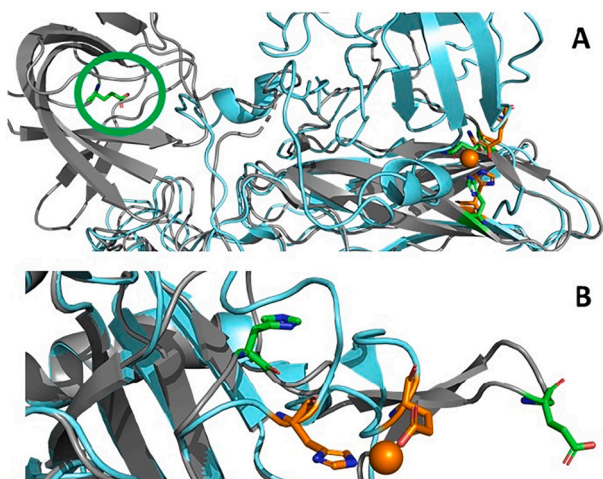


Fig. 7. Examples of large-scale structural rearrangements. A. Apo: 5i5k and Holo: 3cu7 (29.1, 37.4). B. Apo: 2jig and Holo: 3gze (13.9, 18.72). The colour code is as in Fig. 4.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jinorgbio.2022.112025>.

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